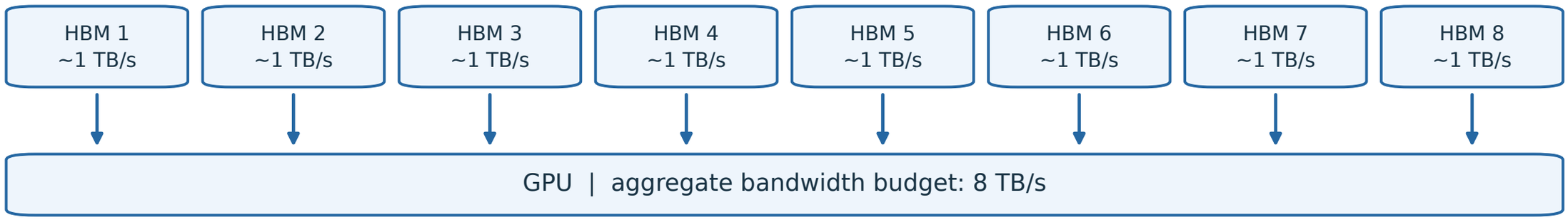
Bandwidth budgets: count independent stacks

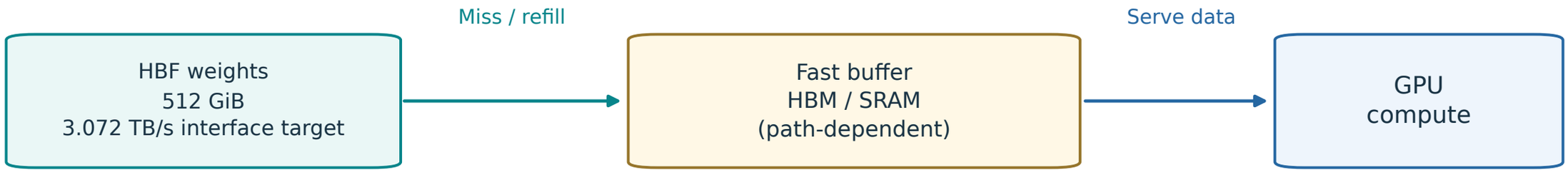
Parallel interfaces can add bandwidth; more NAND die layers do not automatically do so.

A 8-way HBM bandwidth illustration



Equal split for illustration; not a physical package layout or a per-stack product specification.

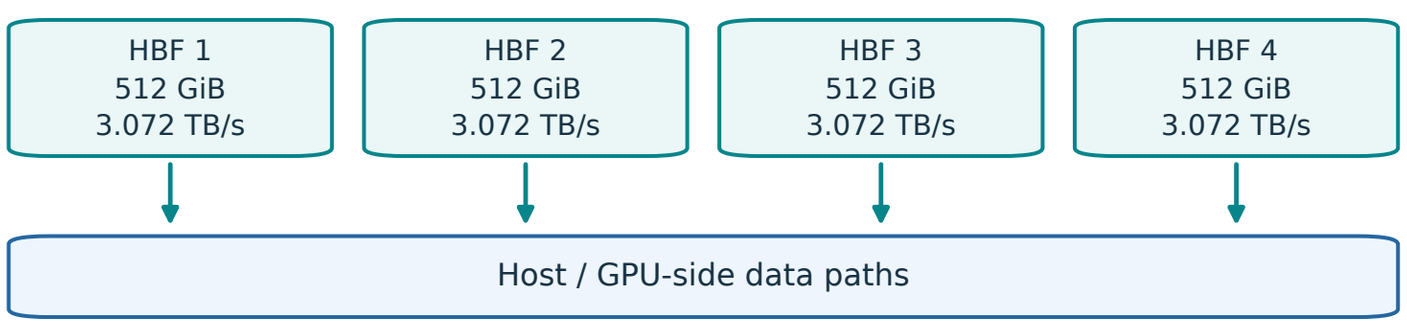
B One HBF source: a fast buffer cannot expand source bandwidth



Cache-miss data from HBF: <= 3.072 TB/s at the source interface.

Hits may reuse resident data. The buffer path is illustrative; HBM staging is not universal.

C Four HBF stacks: a hypothetical sum across independent interfaces



2 TiB capacity

12.288 TB/s

ideal interface sum

Needs host links, balanced data and a viable thermal budget.

Hypothetical bandwidth sum; not a shipping GPU configuration.

Compare parallel stacks. HBF can use multiple stacks; adding layers within one stack mainly adds capacity.